



THE UNIVERSITY
of EDINBURGH

The origins of our Universe

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EIDYN

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MIND + NORMATIVITY

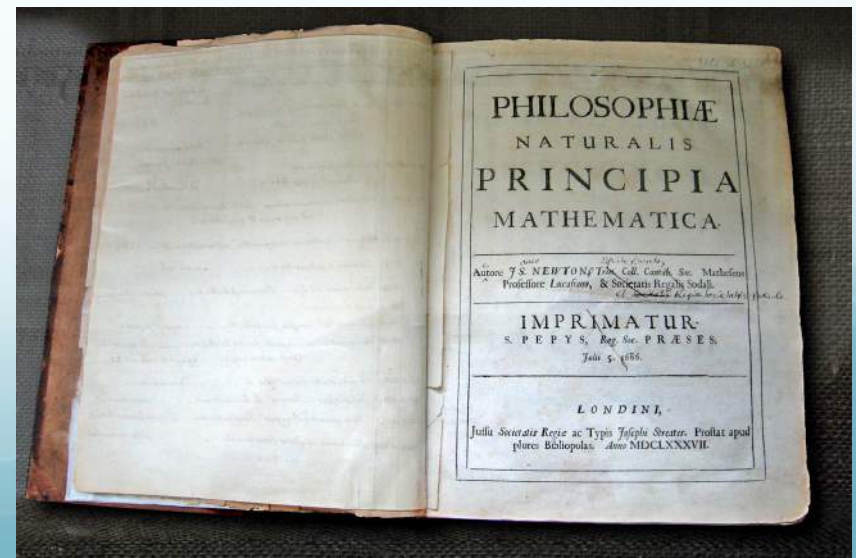
Welcome to Part I of this MOOC

- In the next three lectures, we explore a variety of philosophical issues at the heart of contemporary cosmology.
- Today, we briefly review the history of cosmology, and highlight three main philosophical problems for cosmology.
- In the next class, we focus on two entities at the very center of cosmological research, dark matter and dark energy.
- In the third lecture, we look at the use of anthropic reasoning in cosmology.

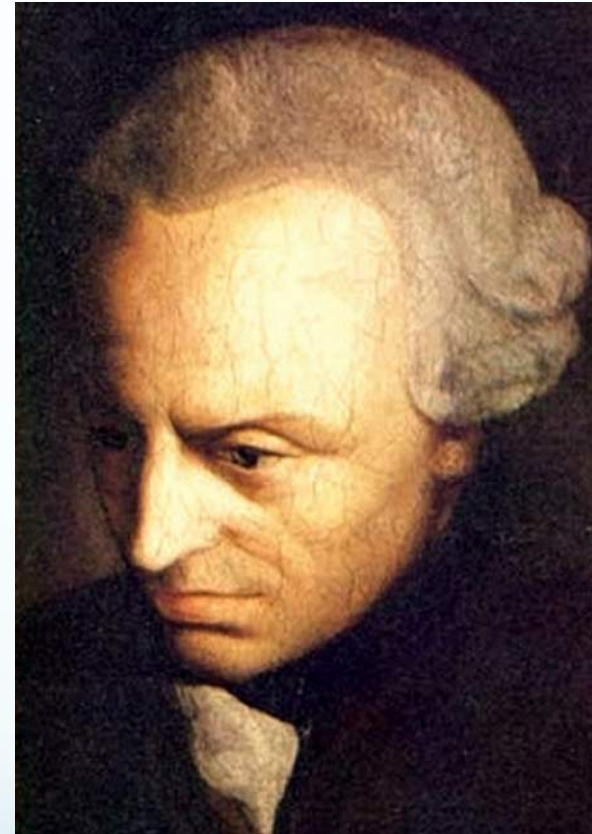
Why cosmology?

What is so special about it?

- Newton's *Mathematical Principles of Natural Philosophy* (1st ed. 1687) provided testable laws that could explain a variety of phenomena.
- Yet Newton's theory did not explain what set planets in motion at the origin of the universe, or how planets and stars formed and how they evolved.
- In the 18th century the task was taken up by philosophers!



- Baumgarten's *Metaphysics* (1739) and Immanuel Kant's *Universal Natural History and Theory of the Heavens* (1755).
- **Kant:** the universe, with planets and stars formed via two opposite forces (attraction and repulsion) acting on a 'fine matter' distributed across all space at the origin.



(Picture: Immanuel Kant. From Wikipedia)



- In this way, Kant anticipated Laplace's **nebular hypothesis**—the hypothesis that the universe formed from an original nebula of gases.
- But why did it take so long for cosmology to become a science in its own right?

(Image: Pierre-Simon Laplace, 1749-1827. From Wikipedia)

Three problems for cosmology

- **1. Laws of nature**

How can we be sure that the same laws we know and love today, apply also to the origins of the universe?

Did not our laws come into existence with the existence of our universe?

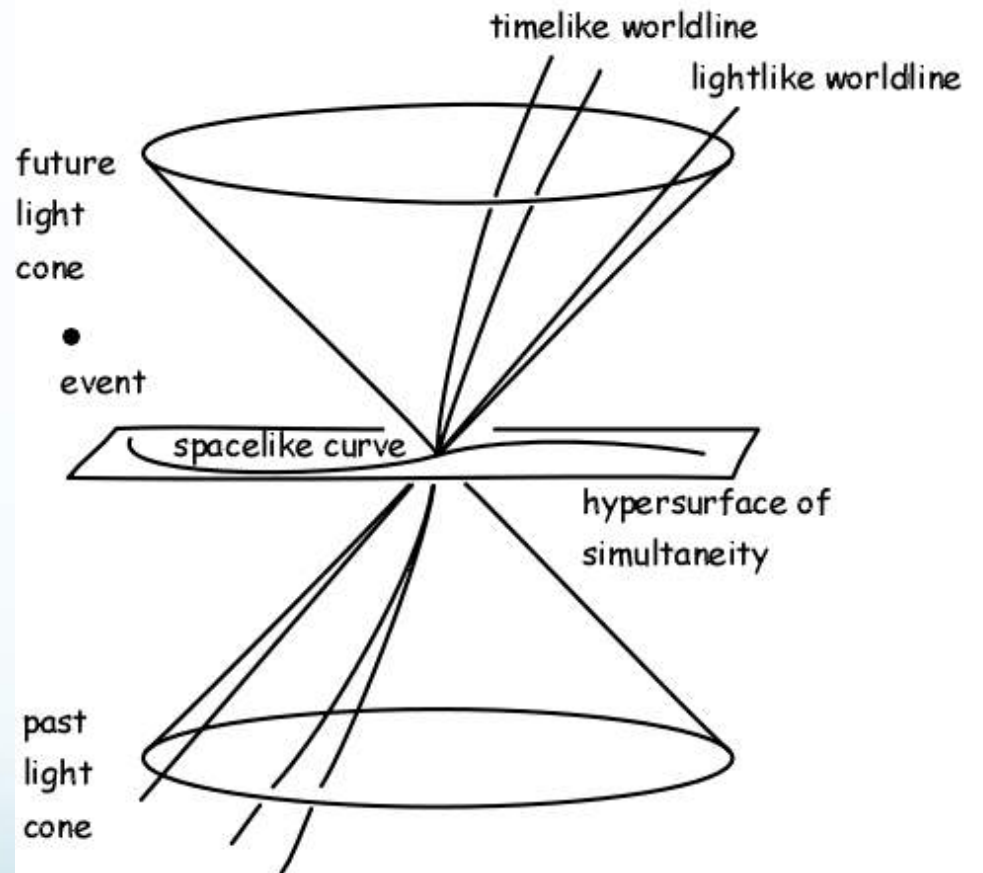
How can we extrapolate from the present physics and its laws, to the physics of the early universe?

- **2. Uniqueness.**

If repeating tests on *multiple samples* and in *different circumstances* is key to experimenting, cosmology faces a problem. We have only one object to observe and to experiment upon: our universe, unique and unrepeatable that it is.

- **3. Unobservability.**

The **past light cone**, i.e. the parts of space that have been able to send information to us given the limit on the velocity of light (300,000 km/sec) constrains what we can ‘observe’, so to speak.

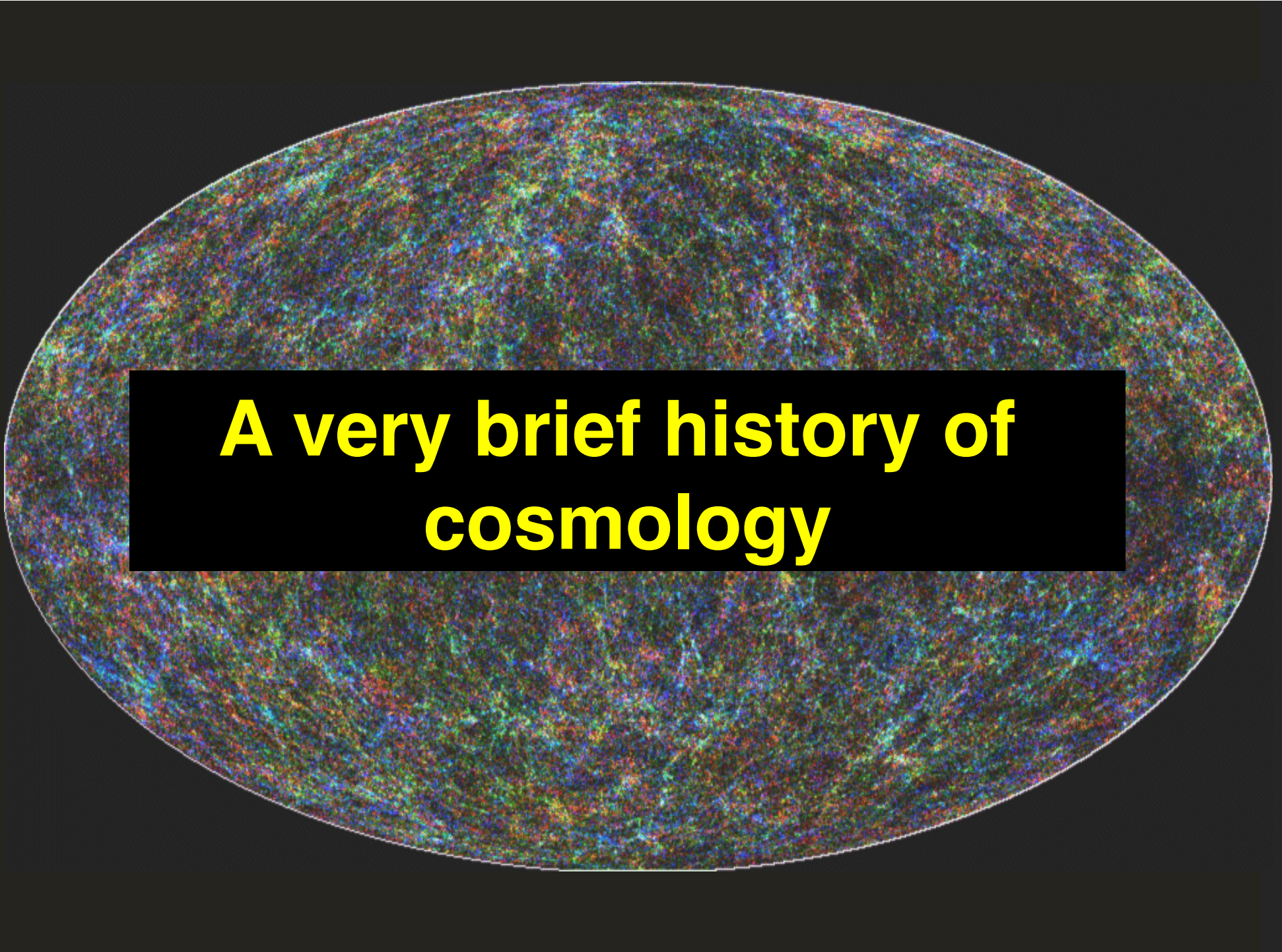


(Image credit: John Norton, “Einstein for Everyone”, open online resource)

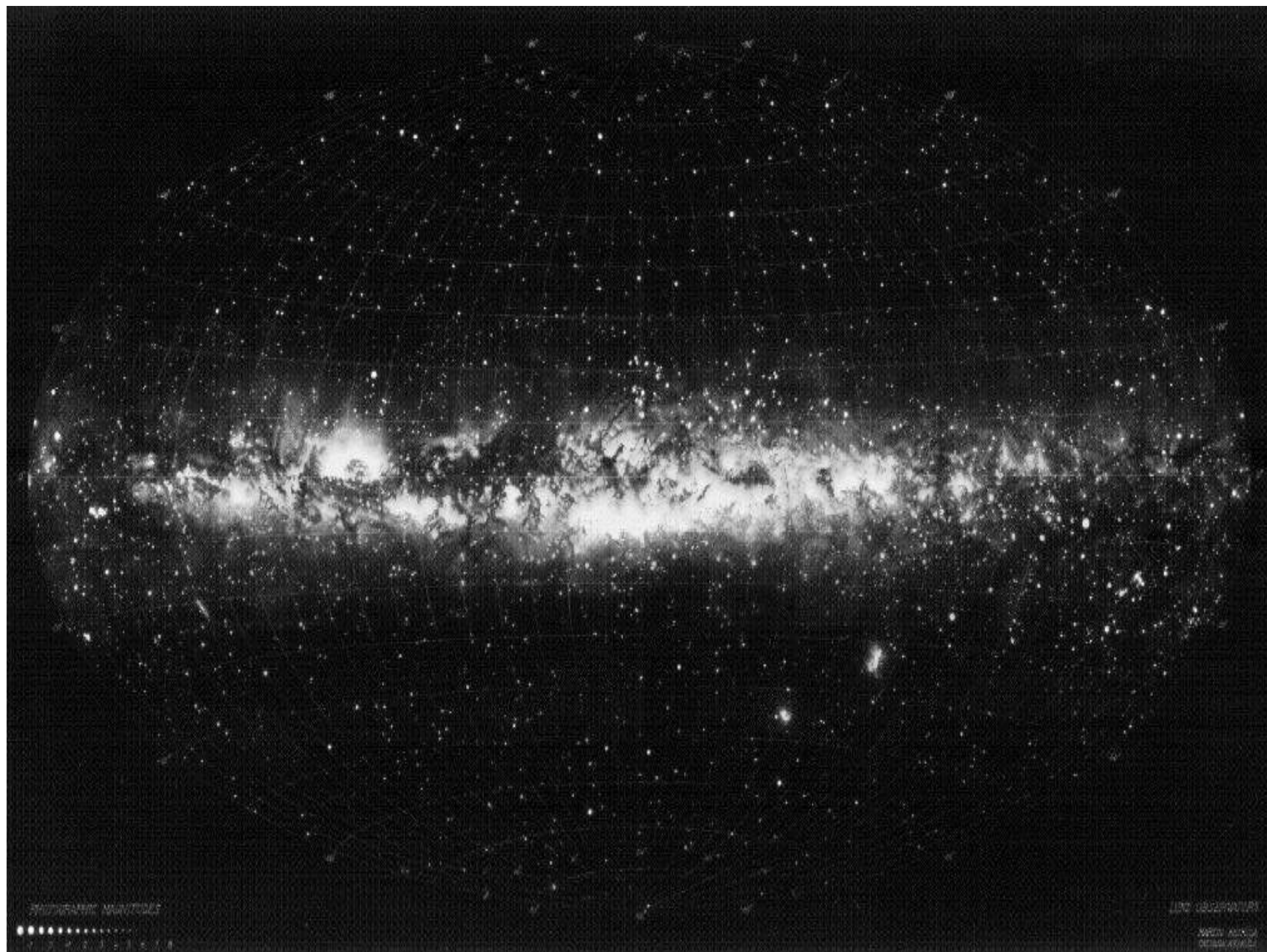
In an accelerating universe, like ours, there exists an **event horizon**: points sufficiently far apart can never contact each other. Thus, there are bound to be vast regions of our Universe that will remain forever unobservable to us.

How can cosmologists make warranted inferences about the universe as a whole?

How could cosmology become a scientific discipline despite these three problems?

A full-page image of a Cosmic Microwave Background (CMB) fluctuation map, showing a noisy, grainy pattern of colors (red, green, blue, yellow) against a black background, representing the early universe's temperature variations. A black rectangular box is centered over the map, containing the title text in yellow.

A very brief history of cosmology



PHOTOMETRIC MAGNITUDES

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1950 OBSERVATIONS

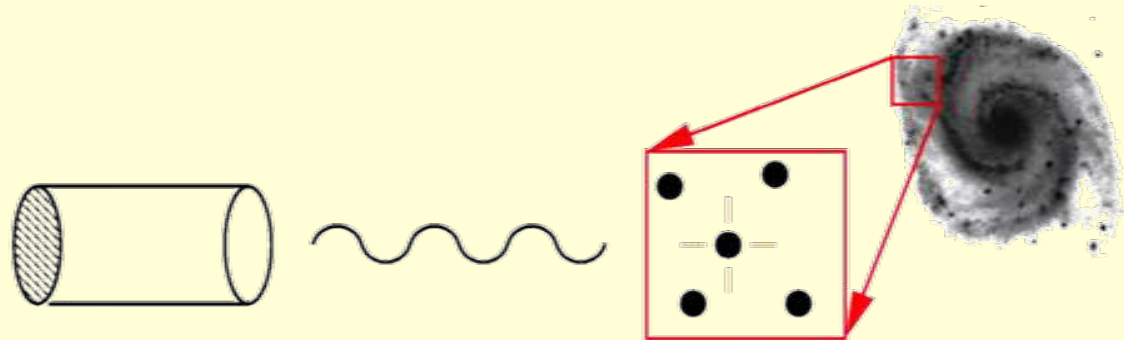
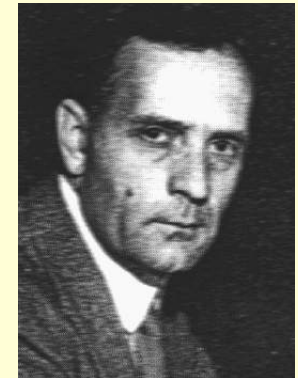
1950 ADONIS
00000000



The Great Debate

Are these clouds of gas, or
distant systems of stars?

1924: Hubble solves the
problem by finding Cepheid
variable stars in M31



1 galaxy = 100 billion stars

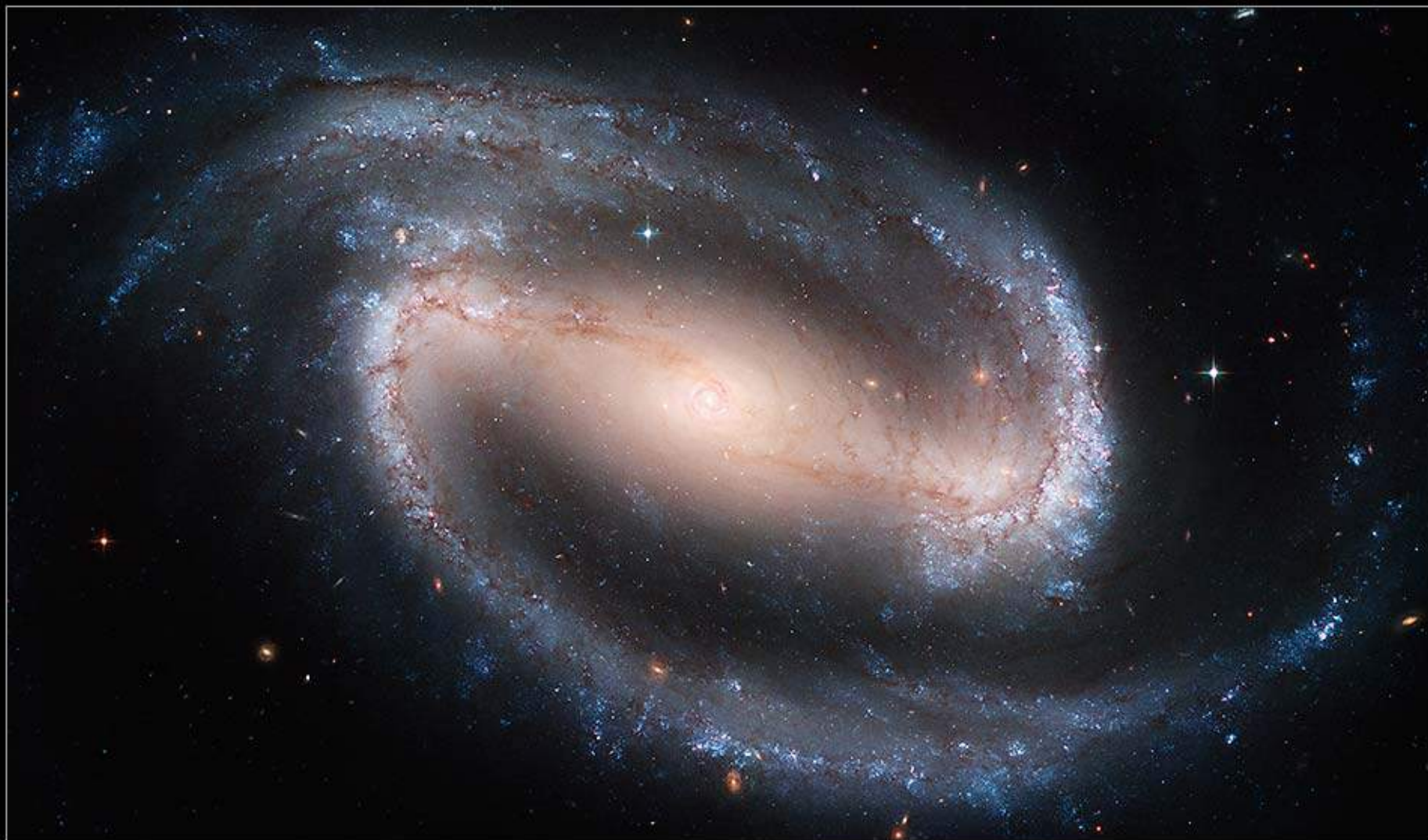
The Hubble Space Telescope



Whirlpool Galaxy · M51



Barred Spiral Galaxy NGC 1300



Galaxy ESO 510-G13

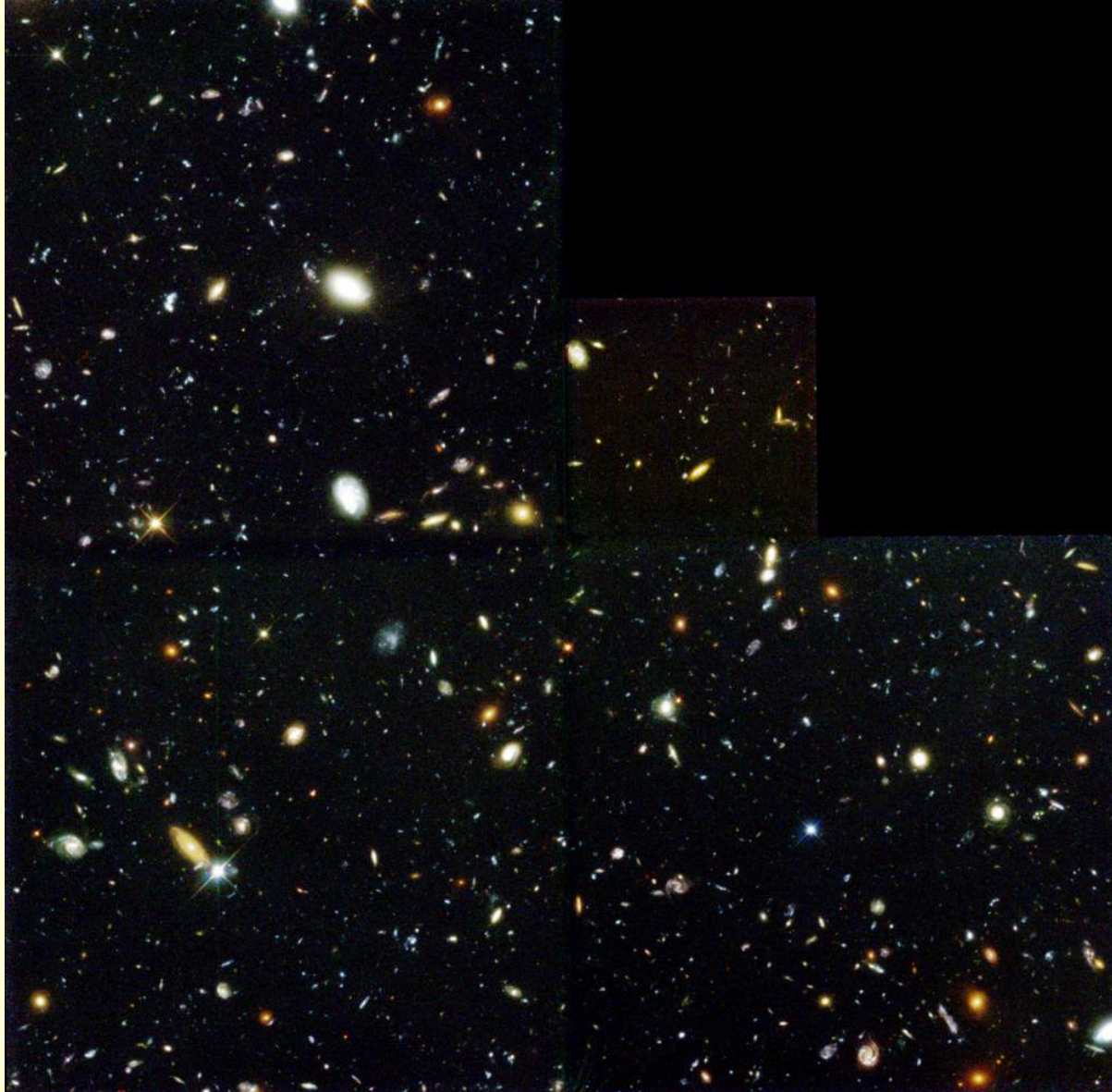


Galaxies NGC 2207 and IC 2163



Hubble
Heritage

The Hubble Deep Field



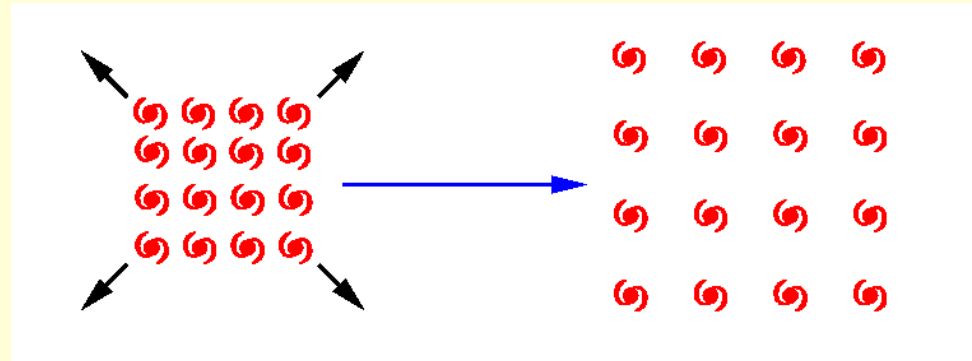
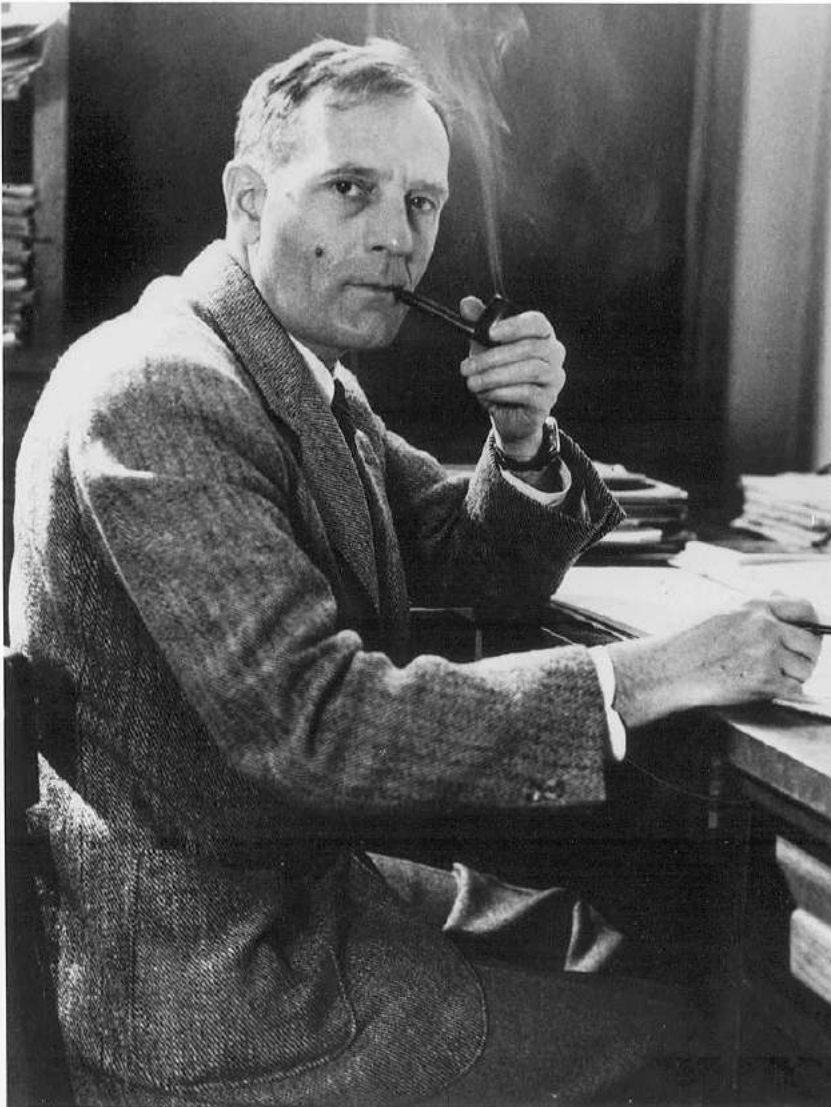
The deepest image
of the sky - about
1% of the moon's
apparent area

Reaches magnitude
28+

- about 1 billion
times fainter than
the human eye

100 billion galaxies
over the whole sky

The Hubble expansion

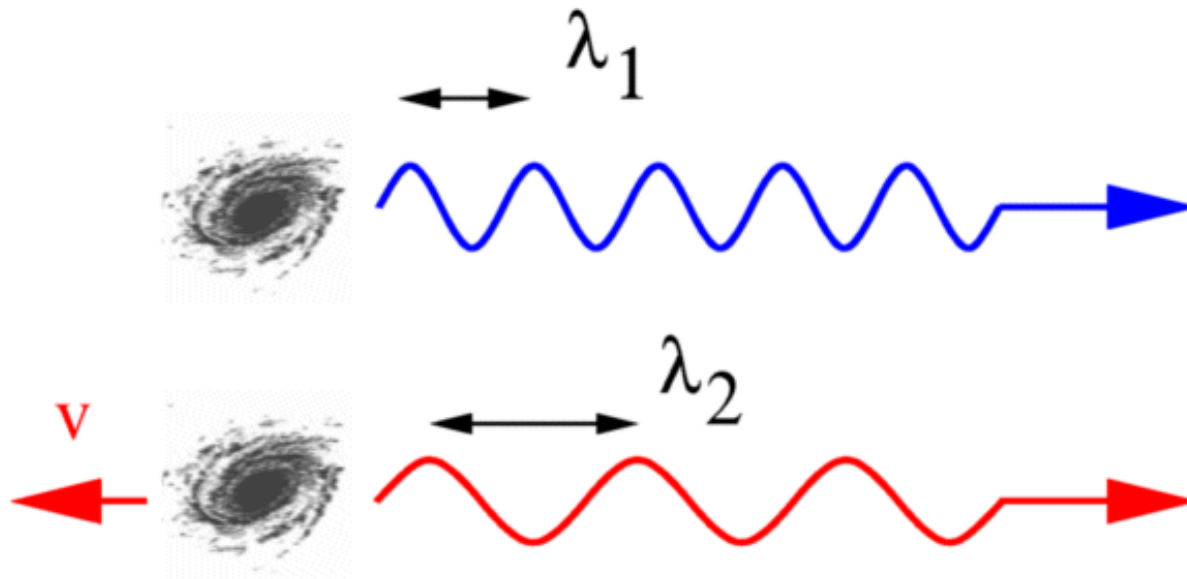


$$V = H D$$

“in 1929, Hubble discovered the expansion of the universe.....”

The expanding universe

Motion of a light emitter stretches/compresses light waves

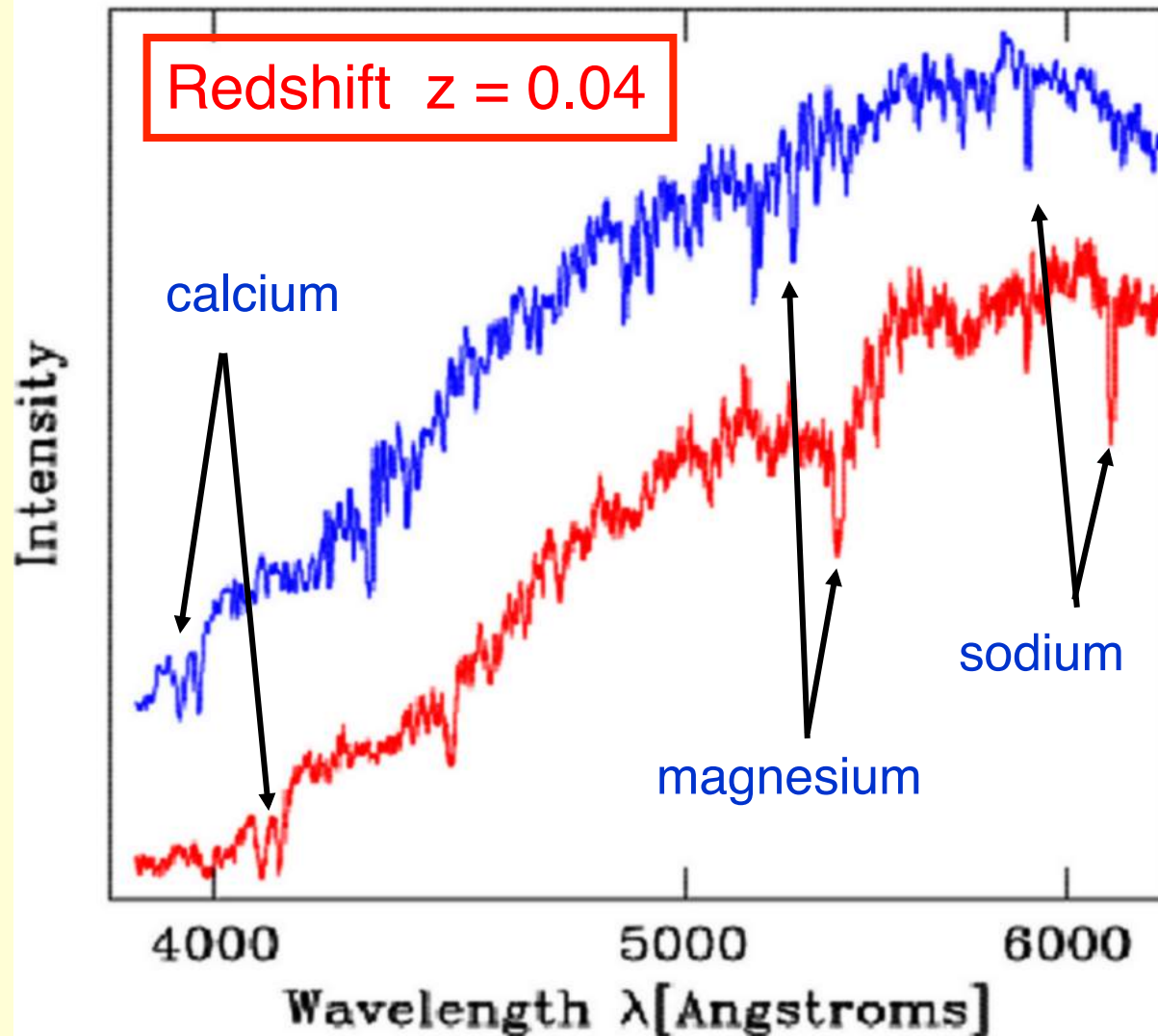


$$1 + z = \frac{\lambda_2}{\lambda_1} \simeq 1 + \frac{v}{c}$$

The Doppler (red)shift



Measuring redshifts

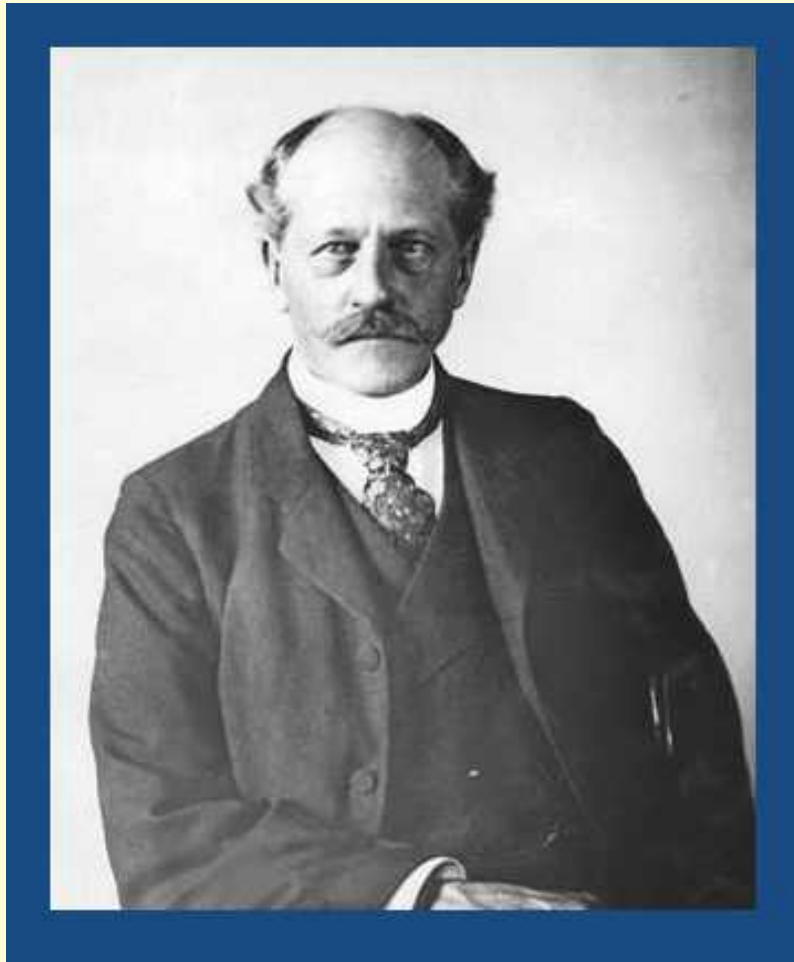


Spectroscopy:

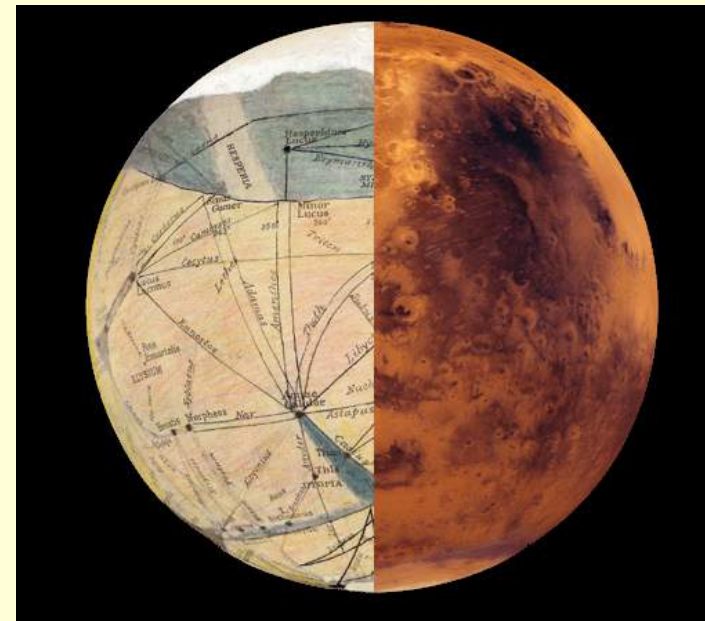
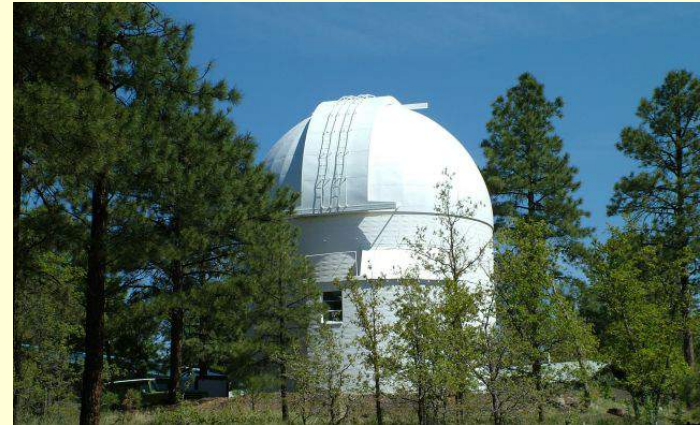
Dispersing the light from galaxies into different colours reveals signatures of atoms as seen on Earth

**So Hubble was the first to
measure redshifts using
spectroscopy?**

The Lowell Observatory, Arizona



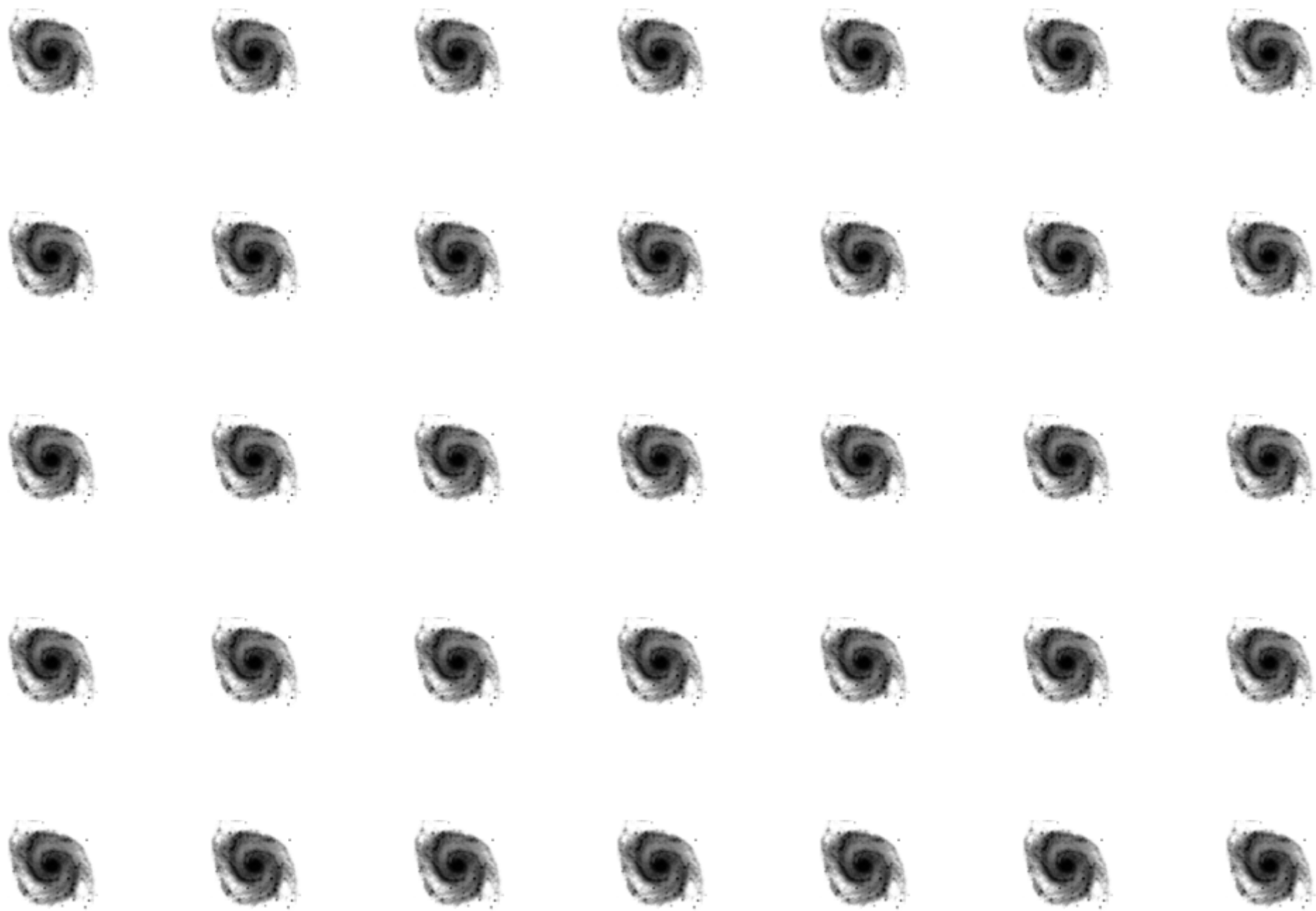
Percival Lowell (1855-1916)



Vesto Melvin Slipher (1875-1969)



- Measured first galaxy redshift on 17 September 1912 (actually a blueshift)
- By 1917, Slipher had spectra for 25 galaxies: 21 had redshifts



Philosophical difficulties with the expanding universe

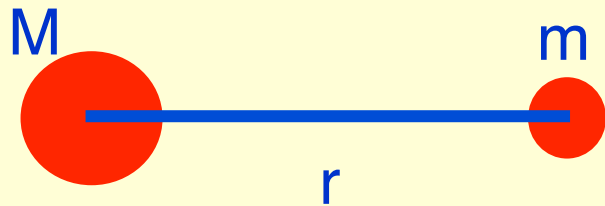
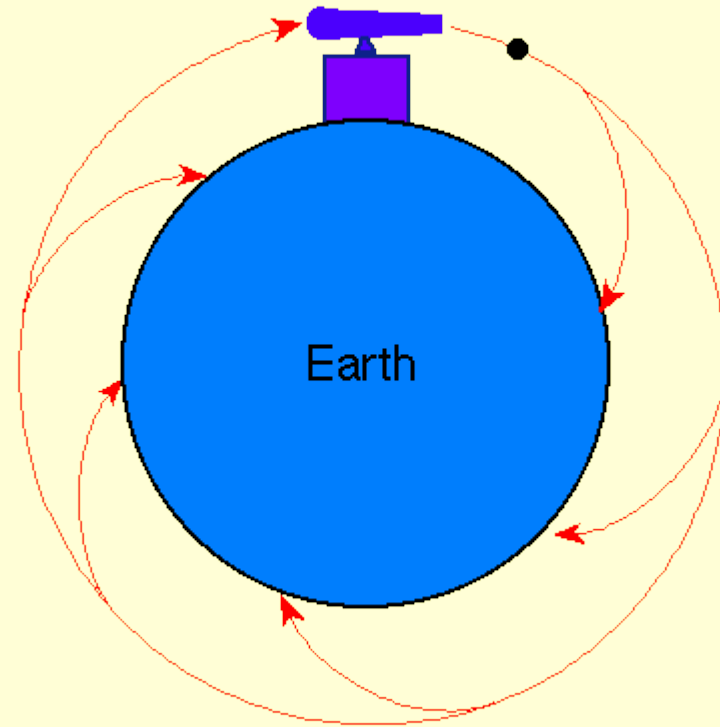
- There is no 'outside' – not expanding into anything
- We are not at the centre – there is no centre
- Space is not 'stretching' locally (a common misconception) – but there is more of it

**Gravity makes (some) sense of
the expansion**

Newton & gravitation



Isaac Newton (1643-1727)

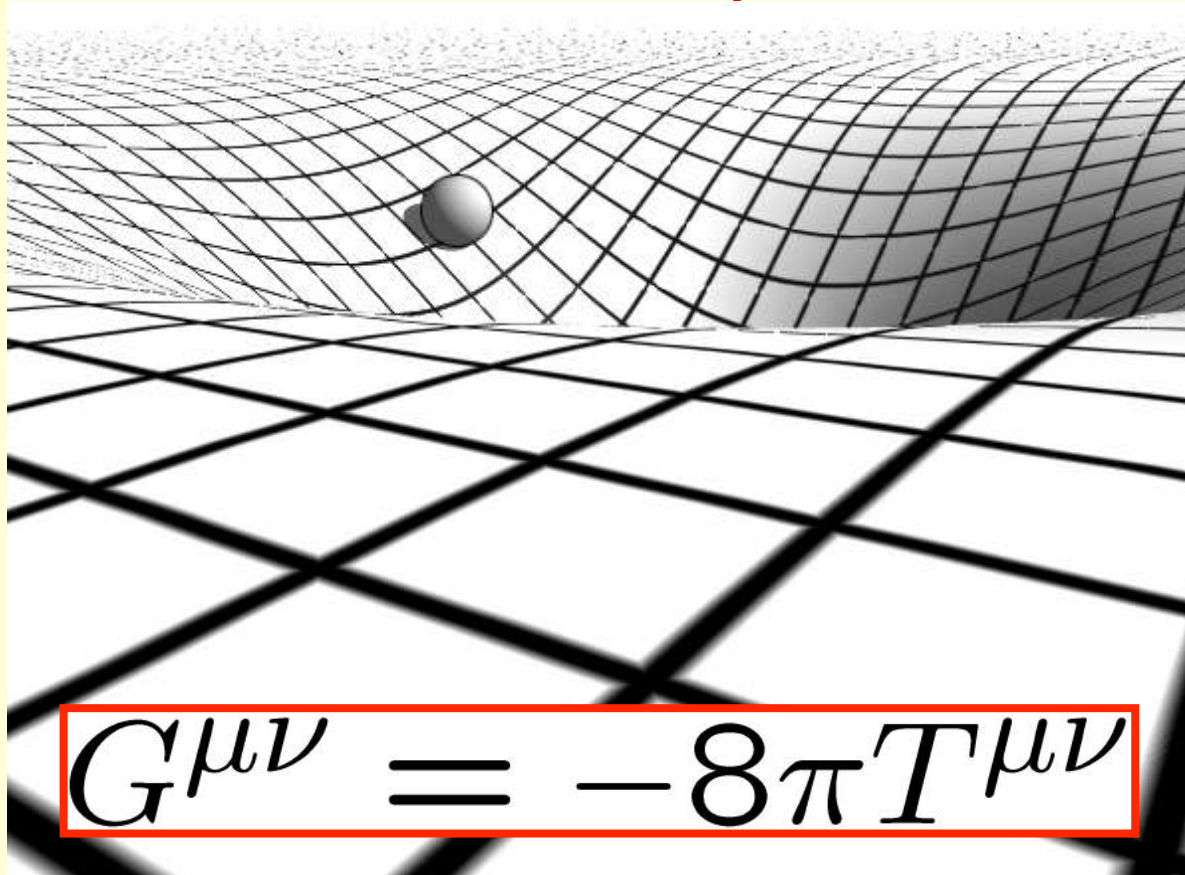


$$F = \frac{GMm}{r^2}$$

Inverse square law:
double distance gives
 $\frac{1}{4}$ the attraction

General Relativity (1915)

Matter bends space-time



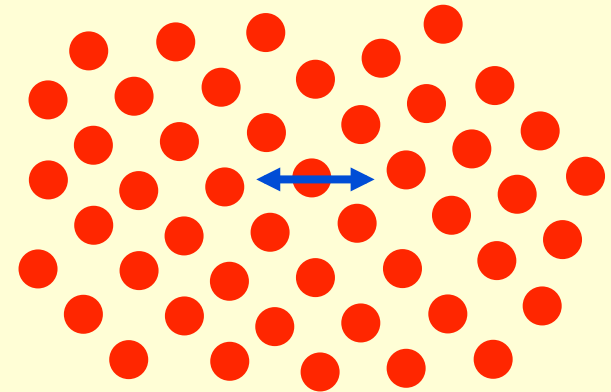
$$G^{\mu\nu} = -8\pi T^{\mu\nu}$$

Matter moves along paths in bent space-time

Newton's missed chance (1692)

Richard Bentley: operation of gravity in an infinite sea of mass?

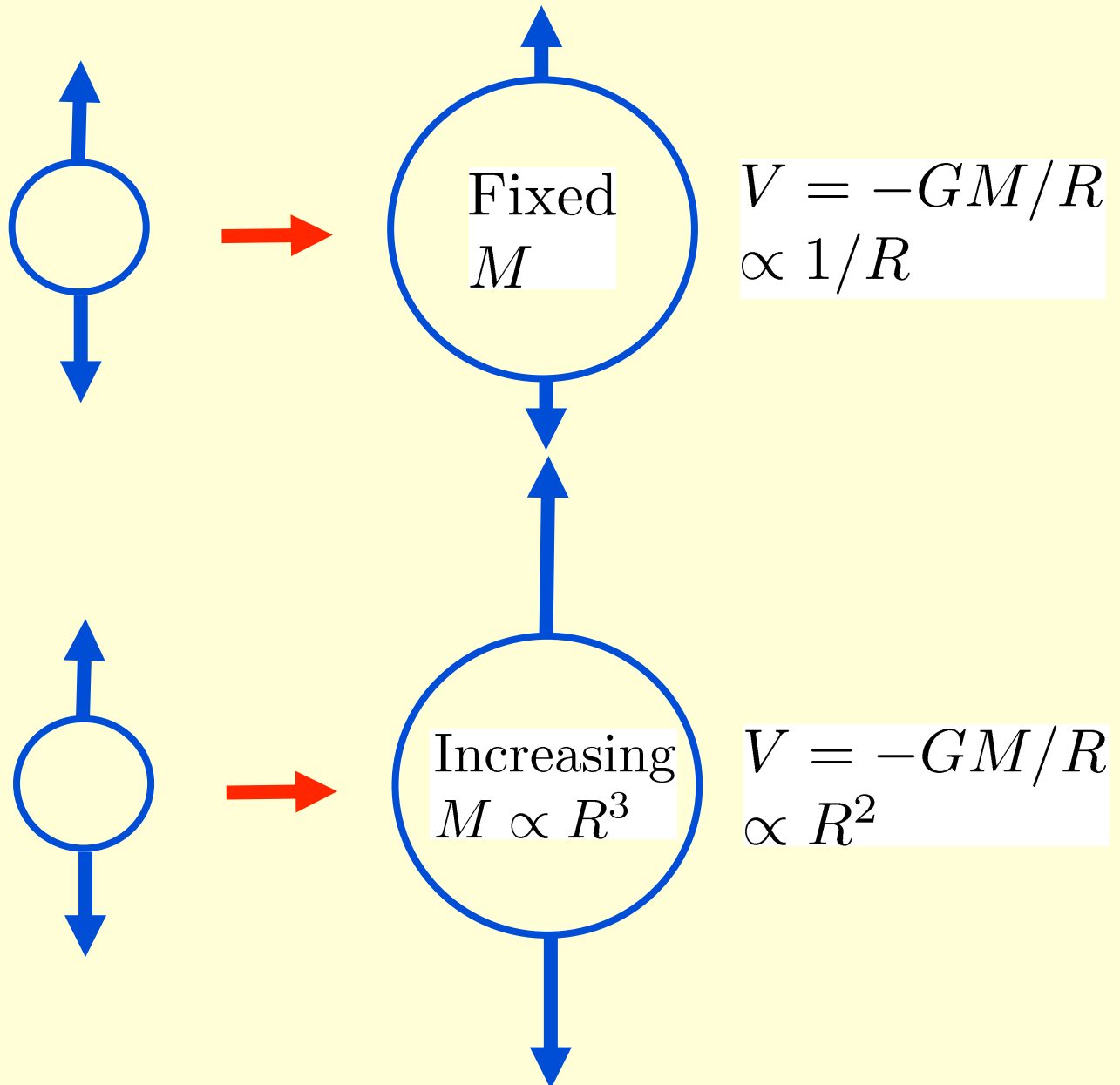
Newton: coalesce into clumps



Einstein (1917): uniform density and zero force aren't consistent even in Newton's gravity. Change law of gravity to introduce 'cosmological constant' – in effect giving a density to empty space

Empty space has antigravity

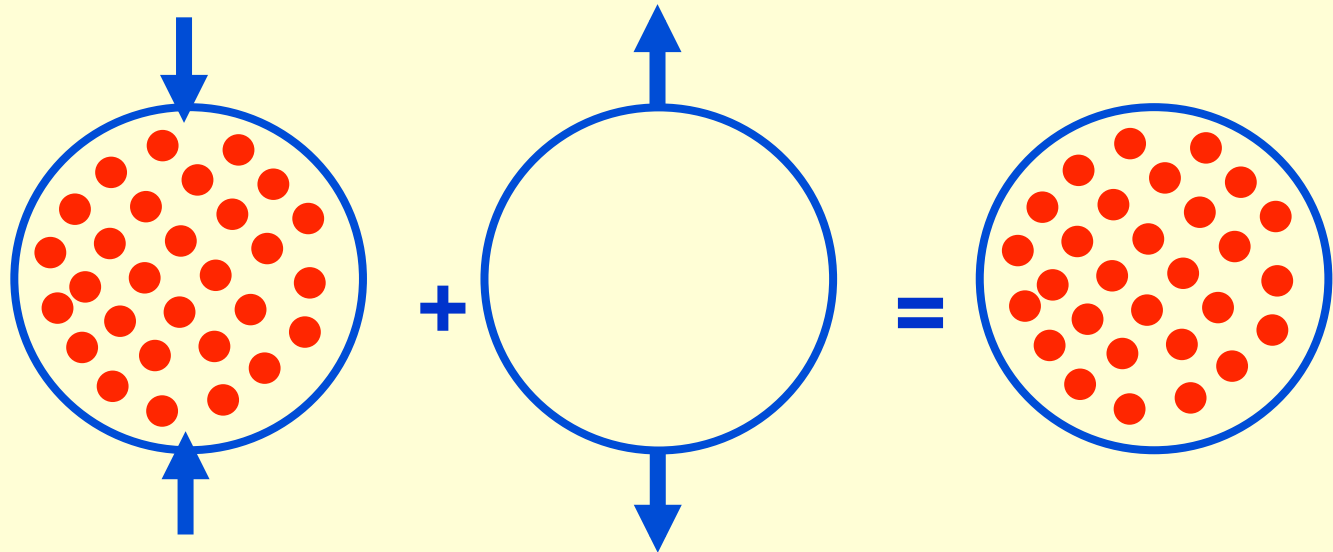
Ordinary matter:
expansion slows
as it expands,
since
gravitational
energy is less



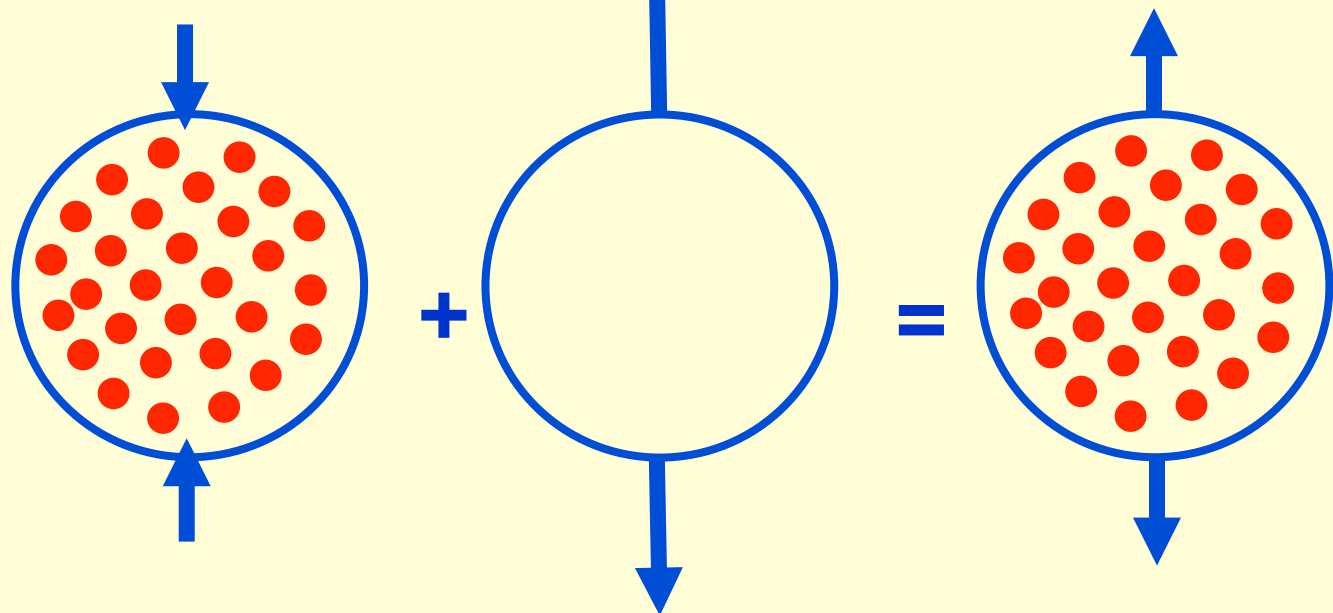
A sphere of
vacuum
increases mass
as it expands,
so gravitational
energy goes up

Vacuum energy: Einstein's missed chance

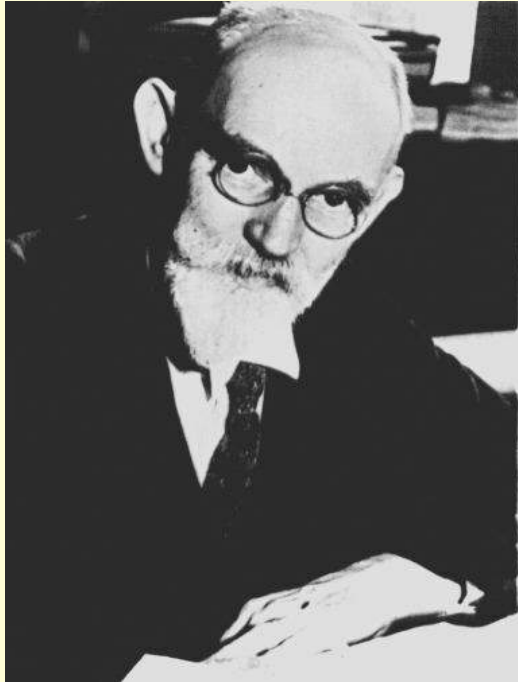
1917: Einstein's static universe balanced gravity and repulsion from **cosmological constant** – abandoned after Hubble



Now: 'Dark Energy' can cause the expansion of the universe to accelerate



De Sitter Space



Willem de Sitter
(1872-1934)

1917: vacuum
dominated relativistic
cosmology



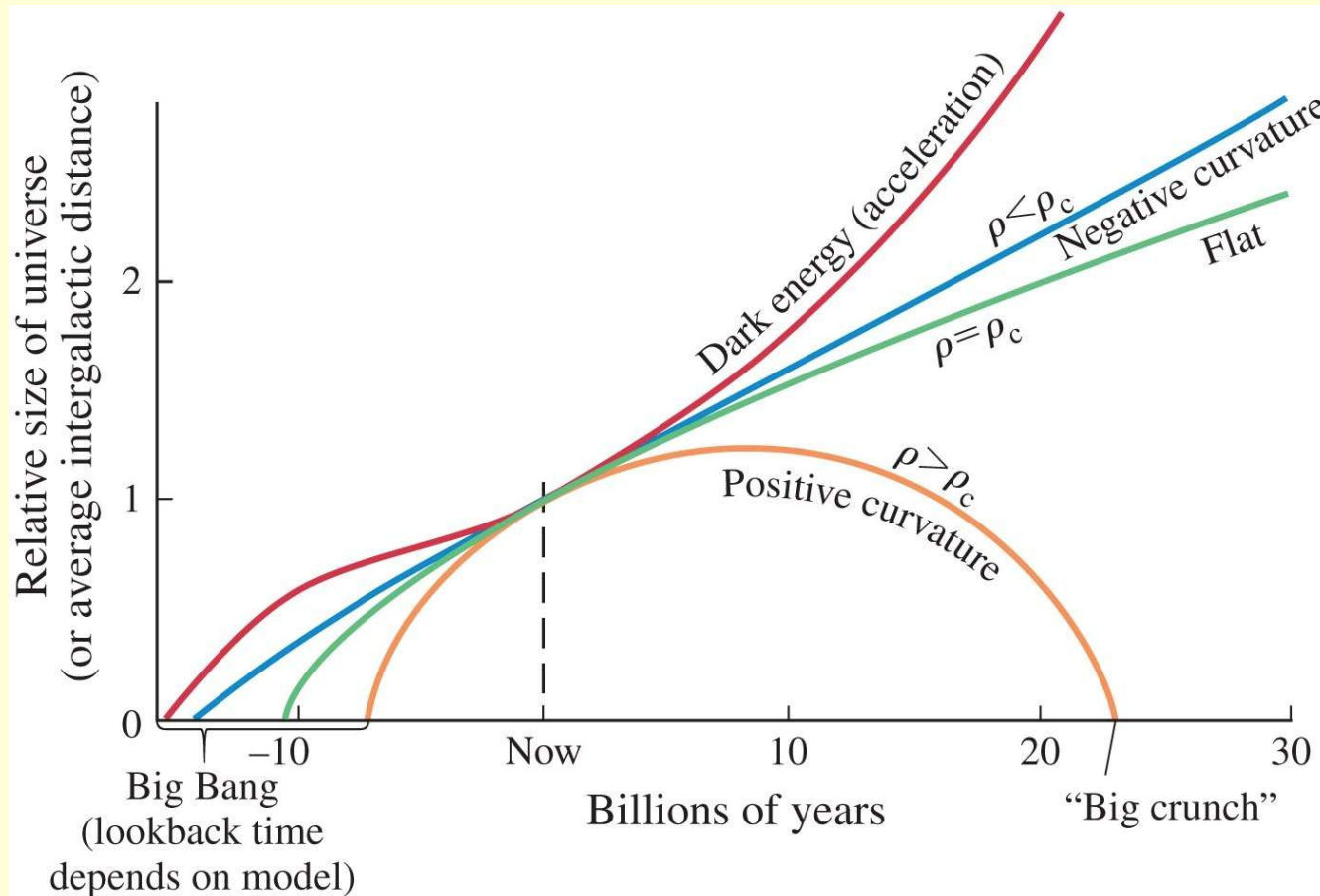
Hermann Weyl
(1885-1955)

1923: expect
linear D - z relation

Many different
searches for
the 'de Sitter
effect' prior to
Hubble 1929

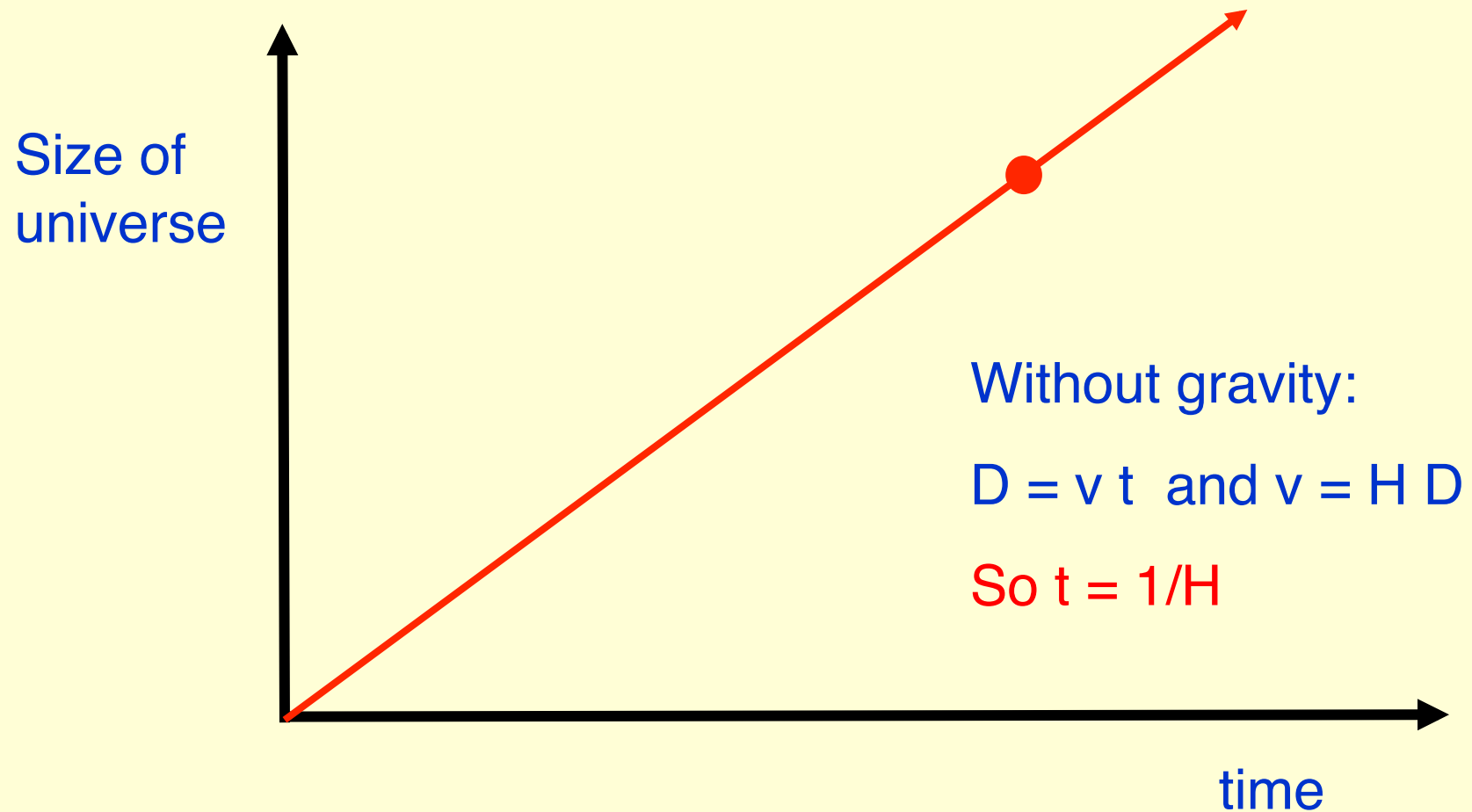
– none thinking
of universe as
expanding

The Big Bang



Friedmann (1922, 1924), Lemaître (1925, 1927): if cosmological constant not too high, expansion always started with infinite density about $1/H$ ago – about 10 Billion years

The Big Bang: birth of the universe?

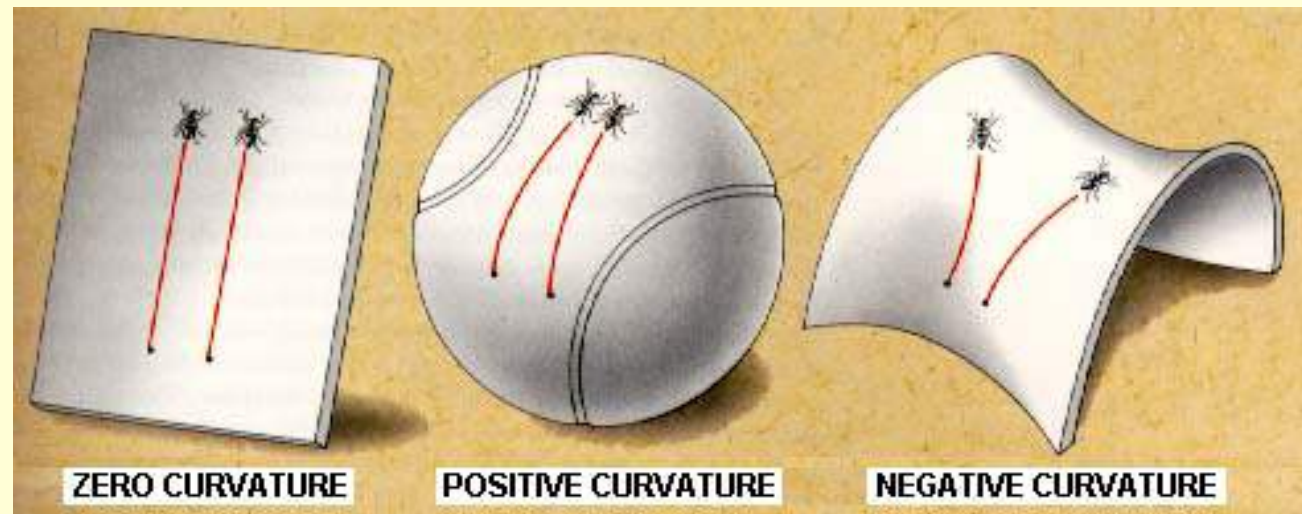


The Big Bang: zero size and infinite density $1/H = 14$ billion years ago

The expanding curved universe

Matter curves
space:

(no outside to
expansion)



flat (infinite)

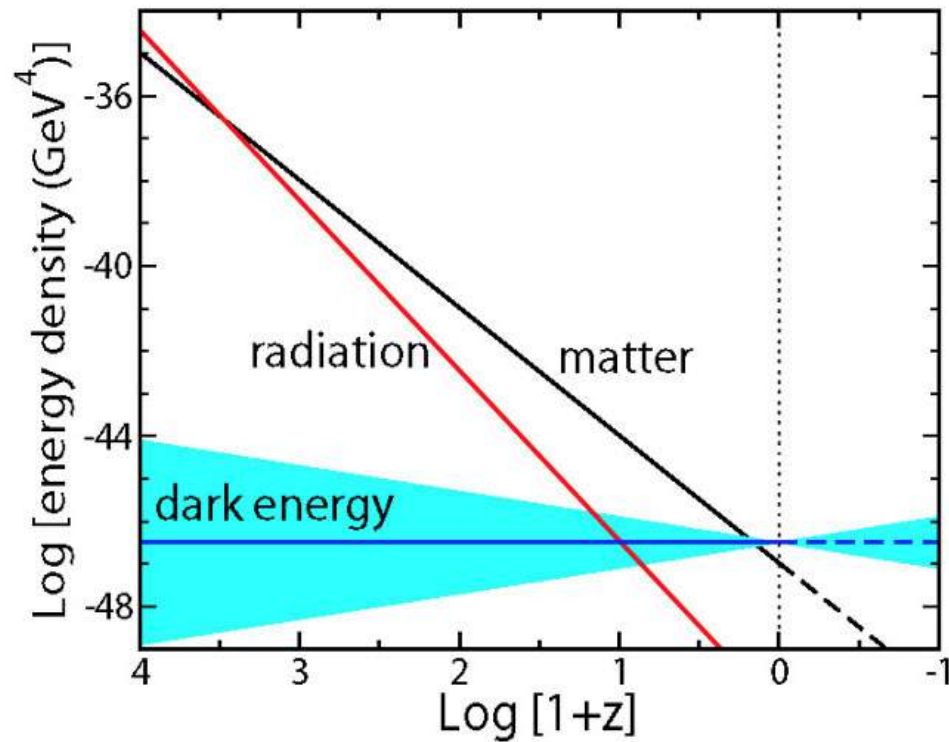
closed (finite)

open (infinite)

‘critical density’ to turn open into closed: about 1 atom per m³

‘density parameter’: $\Omega = (\text{density}) / (\text{critical density})$

The Hot Big Bang



The early universe was small and hot:

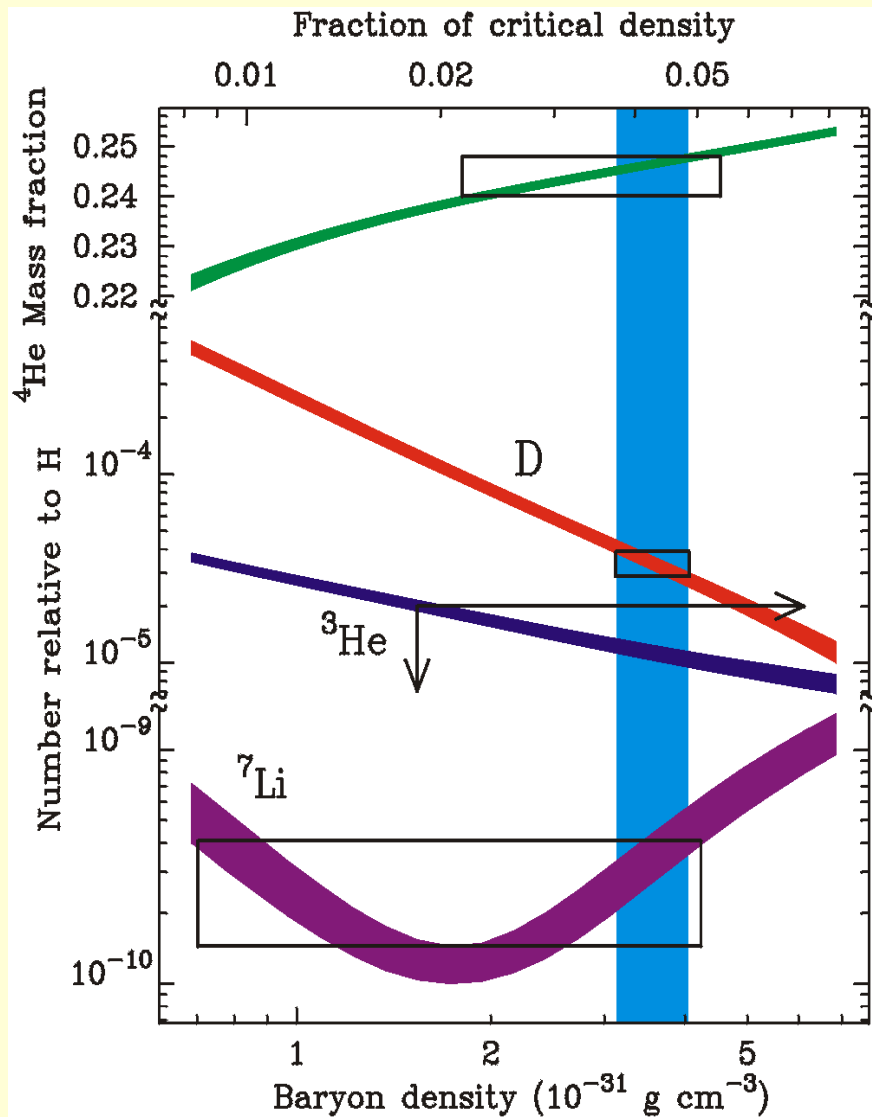
$$T = T_0 (1+z)$$

Redshift factor is ratio of size of universe now and then:

$$1+z = R_0 / R(t)$$

- so energy density of radiation is most important at early times

Nucleosynthesis



D binding E: 2.225 MeV

D forms at $T=10^{8.9}$ K (~ 1 min)

Then He (but D trace remains)

– Explains why all stars have at least 25% Helium

Measure D/H in intergalactic medium $\Rightarrow \Omega_b = 0.045 \pm 10\%$

Great window into exotic physics. E.g. more neutrinos raises relativistic density and changes Helium

$\Rightarrow N(\nu) = 3.1 \pm 0.7$

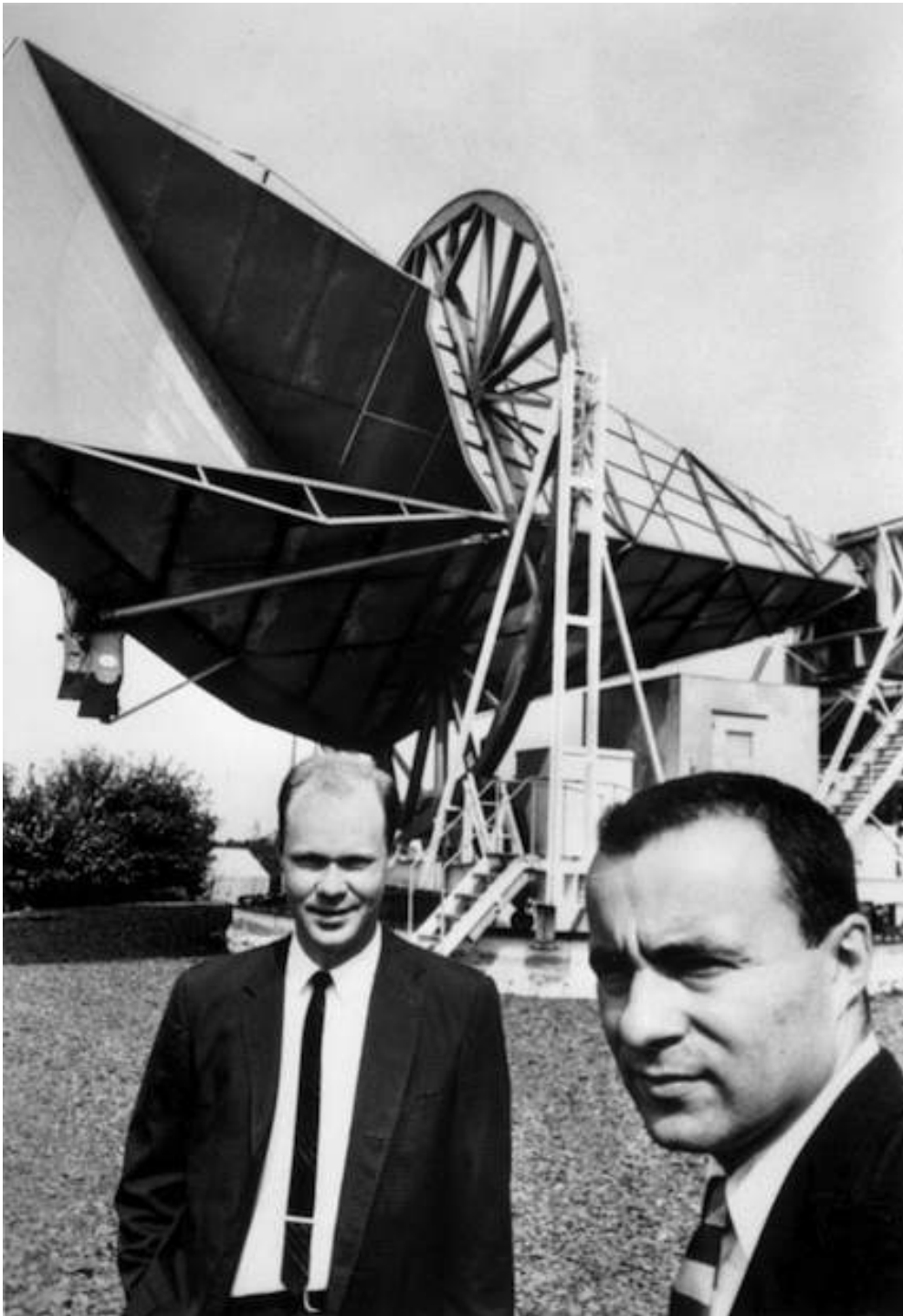
The relic radiation

Alpher, Hermann (1948): know D can form at $T = 1 \text{ MeV}$, but matter density must be high enough then to permit reaction to happen in time. So present matter density gives an estimate of current radiation temperature:



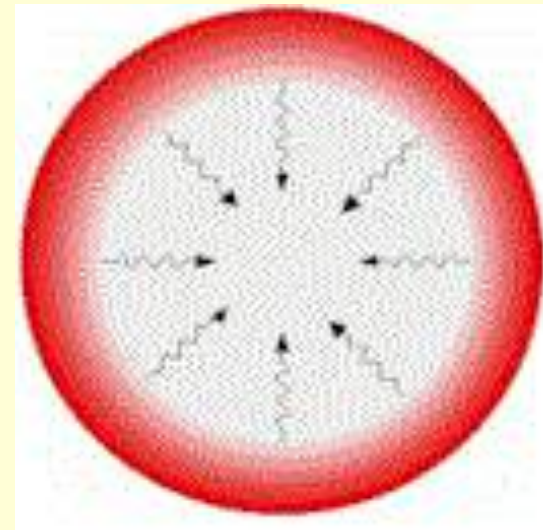
$T \sim 5 \text{ K}$





The Cosmic Microwave Background

$T = 2.7 \text{ K}$ measured by
Penzias & Wilson (1965)



We are surrounded by a
'last scattering surface' at $z = 1100$ ($T = 3000 \text{ K}$)

Cosmology and scientific methodology

- What can philosophers learn from the history of cosmology?
- How does a branch of metaphysics become a scientific theory?
- Let us go back to the three problems

1. Laws of Nature

- The first methodological problem was how to extrapolate the validity of our current laws of nature to the early universe.
- Hermann Bondi and other defenders of the steady-state universe back in the 1950s introduced the so-called **Perfect Cosmological Principle**, which postulates that the universe is homogeneous in its physical laws.
- But the steady-state universe has long been **disproved by experimental evidence** (e.g. CMB) about an evolving universe.

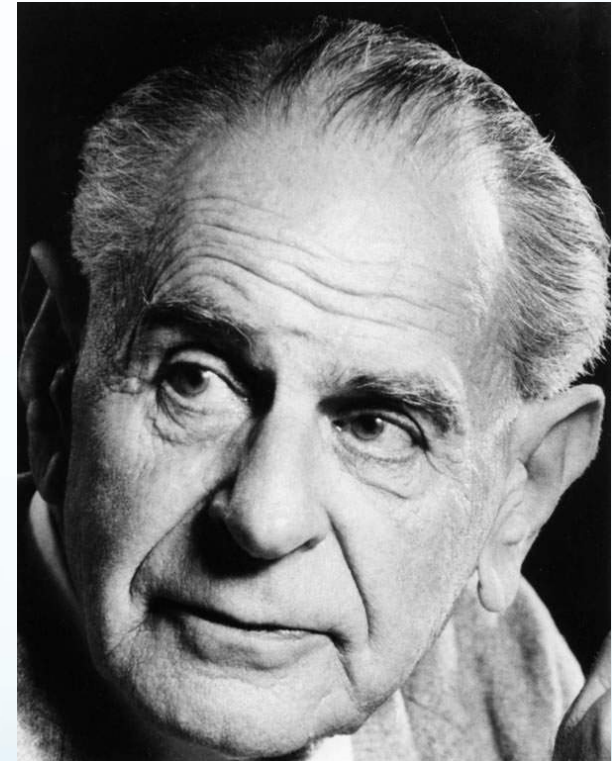


- The physicist Lee Smolin in his latest book *Time Reborn* (2013) defends what he calls the *cosmological natural selection*: laws are not timeless and eternal but instead they evolve in real time

(Image credit: Photo taken by John Norton, Center for the Philosophy of Science, University of Pittsburgh).

2. Uniqueness and Popper's falsificationism

- The second problem with cosmology is the uniqueness of our universe and the specific problem it poses for testability and experimenting.
- Popper's criterion of **falsification** to the rescue? (recall last week's lecture on Popper)



(Image credit: Wikipedia).

- Popper believed that the method of science consisted in a **deductive method**, whereby given a hypothesis or conjecture with **risky novel predictions**, scientists go about and search for one single piece of negative evidence that can potentially falsify the hypothesis.
- All that is needed from cosmology is one single risky prediction, which may be tested and proved wrong (what Popper called a **potential falsifier**).

- Bondi defended Popper's falsificationism as the right method in cosmology
- But Bondi's steady-state universe was itself disproved (via the discovery of CMB) by the same Popperian method he had advocated.
- Please read Helge Kragh's 2013 paper (in the **Additional Resources**) for an overview of how Popper joined these methodological discussions about the falsifiability of cosmology.

3. Observability, indeterminism, and induction

- Recall the third methodological problem posed by cosmology, namely the restricted access to what we can observe about our universe in terms of the past-light cone of Earth-now.
- The main problem here is a form of **indeterminism about spacetime**.

- There might be **observationally indistinguishable spacetimes** (i.e. different models of spacetime, which are nonetheless compatible with the same observable past light cone of events).
- Hence given our past light cone, we might inductively infer to very different, and yet observationally indistinguishable spacetime models, and no facts can make the inference to one of these models more legitimate or justified than to another (please read John Norton's article in the **Additional Resources**).

Summary

- For centuries cosmology was regarded as a branch of metaphysics rather than a science.
- The Kant–Laplace nebular hypothesis is one of the first attempts at a scientific explanation of the origins of the universe.
- Cosmology faces three distinctive methodological problems as a science: the applicability of the laws of nature, the uniqueness of its object of study, and the unobservability of vast portions of the universe.

- Progress in the subject has then depended on the large-scale uniformity of the universe, and observational probing of the distant universe via Hubble's law, where the spectroscopic redshift is linearly proportional to distance.
- Theory played a large part in establishing cosmology. de Sitter found the first expanding cosmological model, predicting Hubble's law.
- Generalization by Friedmann showed that the origin of the expansion lay in a past singularity — the Big Bang. With the discovery of the Cosmic Microwave Background in 1965, this hot origin for the universe became the accepted picture.

- The main constituents of the universe seem to be *dark matter* and *dark energy*. The former is a form of matter that can clump; the latter is an energy density associated with empty space, which causes the expansion of the universe to accelerate.
- The big problems in cosmology are to understand what these dark components are, and also why the universe contains small fluctuations in density in order to seed the future formation of galaxies and other astronomical structures.
- Possible ways of approaching such issues in cosmology are either by revising our philosophical intuitions about what laws of nature are and how they can be tested; or by assuming that the laws may evolve and change over time with our universe.

- Despite there being only one object to study, our universe, cosmology seems amenable to Popper's method of falsification, whereby evidence coming from cosmic microwave background eventually falsified the hypothesis of the steady state universe, for example.
- The restriction of observability to the past light cone of events means that there might well be many observationally indistinguishable spacetimes and any inductive inference from available data to the exact nature of the spacetime we live in, is inevitably unwarranted.

THANK YOU FOR LISTENING!